

命题逻辑

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学习内容

命题逻辑的表示:

- A. 基于知识库的Agent(KBA)
- B. Wumpus世界
- C. 逻辑/命题逻辑的语法、语义
- D. 模型、有效性、可满足性
- E. 蕴涵与逻辑等价
- F. 逻辑推理

逻辑表示
逻辑推理

A. 基于知识的AGENT(KBA)



基于知识的Agent

- 知识从程序中分离出来
 - 基于知识的Agent(KBA)
- 知识如何表示
 - 逻辑作为支持KBA的一类通用表示
 - 陈述性知识

• 过程性知识
在前面的问题求解Agent中，
知识隐藏在特定领域的Result
函数代码中。

成功的Agent在设计中必须将陈述性和过程性的成分相结合，而且陈述性知识通常被编译成更有效的过程代码。

- KBA如何进行推理
 - 归结推理是方法之一

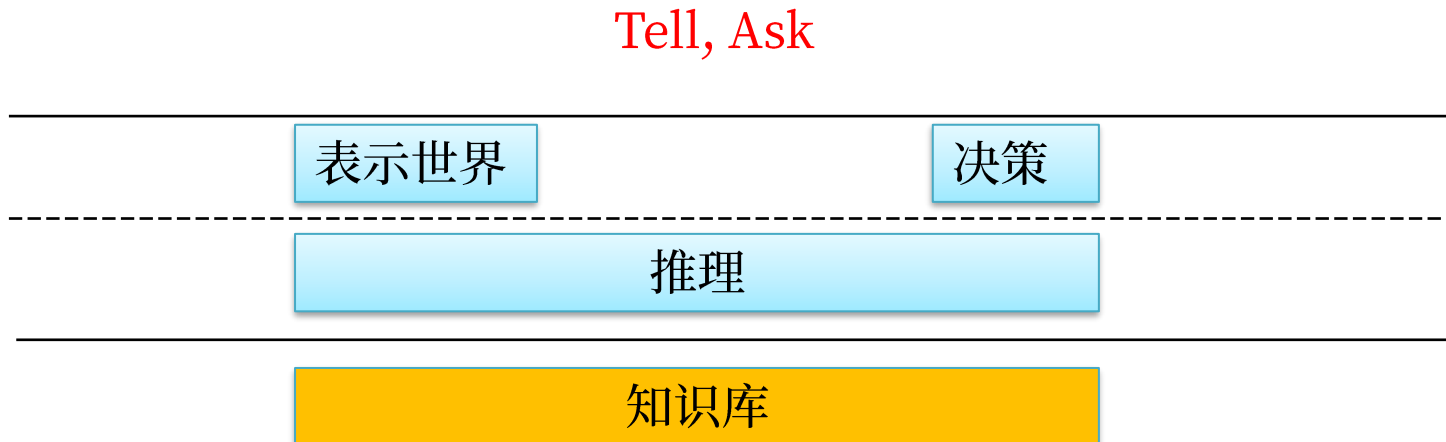
KBA的核心部件

- KBA的核心部件是知识库。知识库是一个语句集合。语句用知识表示语言表达，表示关于世界的断言。直接给定的语句是公理，推导得到的语句是定理。



Tell和Ask

- Two operations on KBs
 - Tell and Ask
 - Both may involve inferencing(推理), deriving new sentences from old

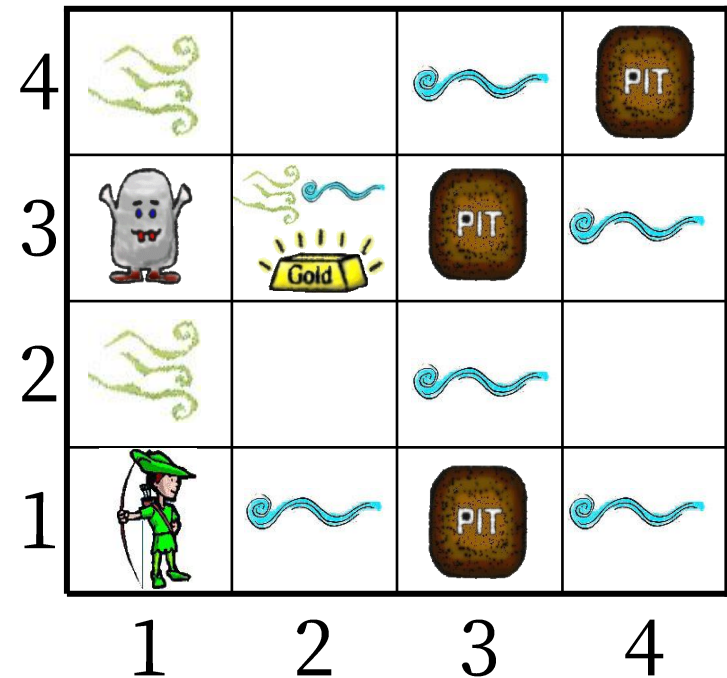


B. WUMPUS世界



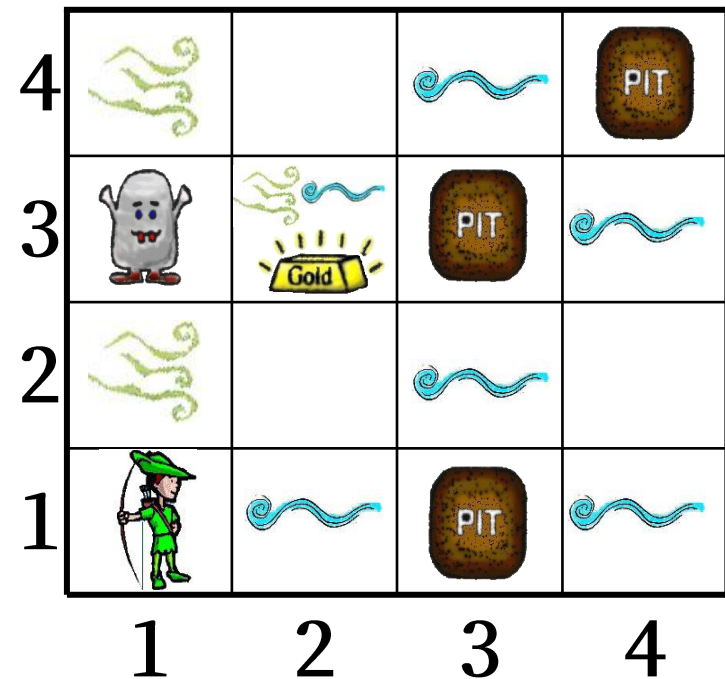
Wumpus世界

- The **wumpus world** is a cave consisting of rooms connected by passageways. Lurking somewhere is the **wumpus**, which eats anyone who enters its room. The wumpus can be shot by **an agent**, but the agent has only **one arrow**. Some rooms contain **pits** that will trap anyone who wanders into these rooms (except for the wumpus, which is too big). The only mitigating feature of this bleak environment is the possibility of finding a heap of **gold**.
- What is the PEAS of this world?
- On-line? Off-line?



Wumpus世界的PEAS

- **Performance measure:** gold +1000, death -1000, -1 per step, -10 for using the arrow
- **Environment:** A 4×4 grid of rooms. Start at [1,1], facing to the right. The locations of the gold and the wumpus are chosen randomly (and uniformly) from the squares other than the start square. In addition, each square other than the start can be a pit, with probability 0.2.



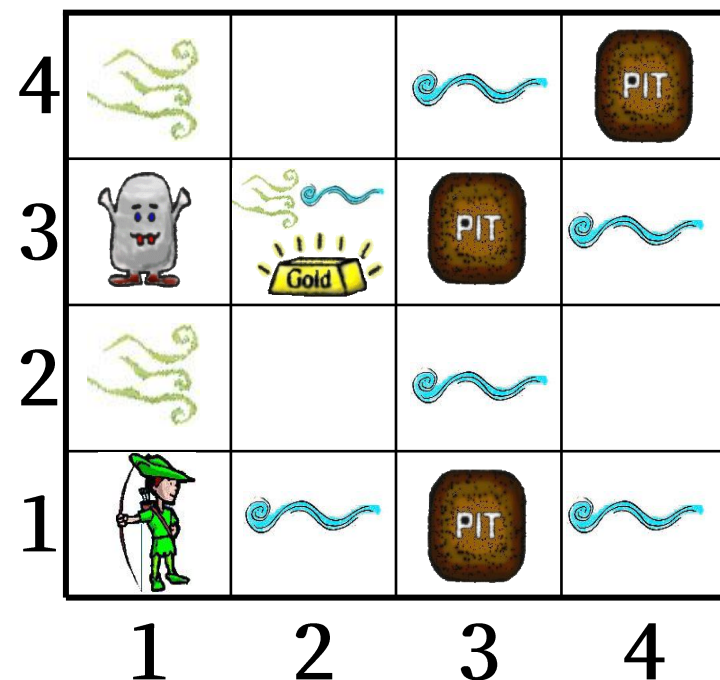
Wumpus世界的PEAS

□ Actuators

- Left turn
- Right turn
- Forward
- Grab
- Shoot

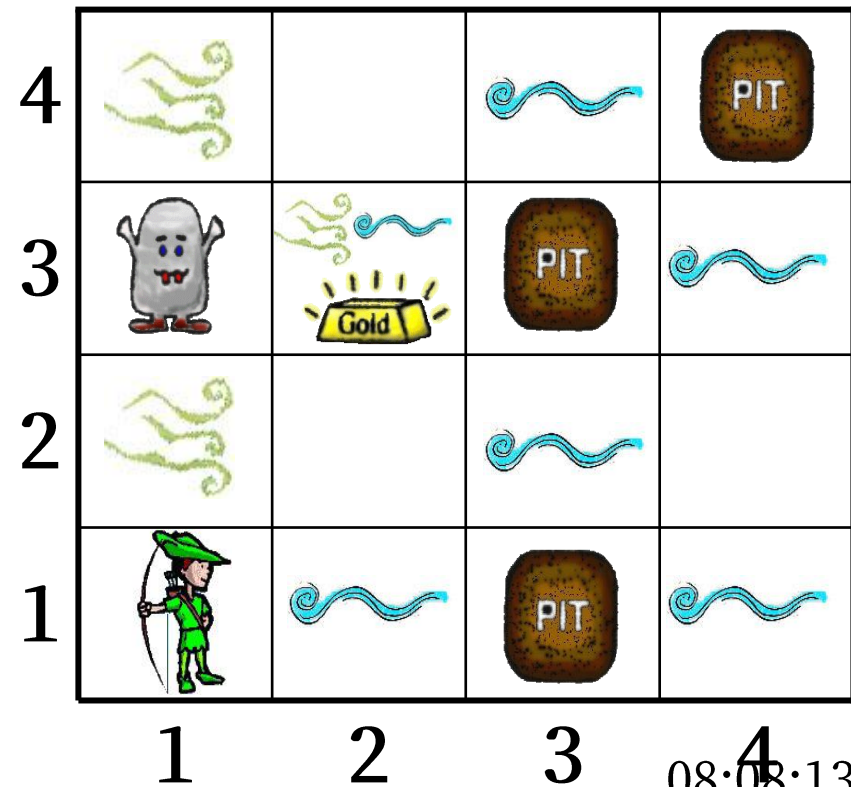
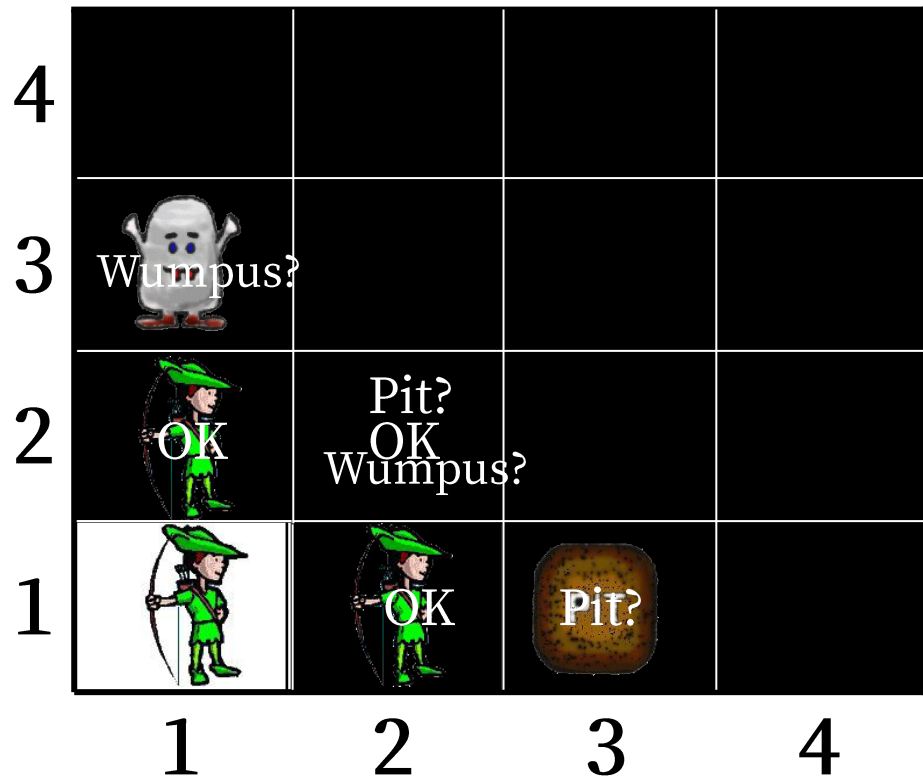
□ Sensors

- Stench(臭气), in, 4-adjacent
- Breeze(微风), 4-adjacent
- Glitter(闪光), in
- Bump, walking into a wall
- Scream, wumpus killed, anywhere



Wumpus世界的环境探索

- ☐ Agent needs to know which actions are safe.
- ☐ reasoning
- ☐ Wumpus world can be solved using logic.



C. 逻辑的语法和语义



逻辑的语法

- **Logics** are formal languages for representing information such that conclusions can be drawn
- **Syntax** defines the form of sentences
(为合法语句给出规范)
- E.g., the language of arithmetic
 - $x+2 \geq y$ is a sentence; $x^2+y > \{\}$ is not a sentence
- E.g., propositional logic
 - $P \wedge Q$ is a sentence; $\wedge Q$ is not a sentence

命题逻辑的语法

- Propositional logic (命题逻辑) is the simplest logic – illustrates basic ideas
- The **proposition symbols** P_1, P_2 etc are sentences
 - If S is a sentence, $\neg S$ is a sentence
(**negation**, 否定式)
 - If S_1 and S_2 are sentences, $S_1 \wedge S_2$ is a sentence
(**conjunction**, 合取式)
 - If S_1 and S_2 are sentences, $S_1 \vee S_2$ is a sentence
(**disjunction**, 析取式)
 - If S_1 and S_2 are sentences, $S_1 \Rightarrow S_2$ is a sentence
(**implication**, 蕴含式)
 - If S_1 and S_2 are sentences, $S_1 \Leftrightarrow S_2$ is a sentence
(**biconditional**, 双向蕴含式)

逻辑的语义

- **Semantics** defines the truth of each sentence w.r.t. each possible world
(定义每个语句在每个可能世界的真值)

- E.g., the language of arithmetic

- $x+2 \geq y$ is true in a world where $x=7, y=1$
- $x+2 \geq y$ is false in a world where $x=0, y=6$

变量的赋值

- E.g., propositional logic

- $P \wedge Q$ is true in a world where $P=T, Q=T$.

命题符号的赋值

命题逻辑的语义

- Rules for evaluating truth with respect to model m
 - $\neg S$ is true iff S is false
 - $S_1 \wedge S_2$ is true iff S_1 is true and S_2 is true
 - $S_1 \vee S_2$ is true iff S_1 is true or S_2 is true
 - $S_1 \Rightarrow S_2$ is true iff S_1 is false or S_2 is true
 - $S_1 \Leftrightarrow S_2$ is true iff $S_1 \Rightarrow S_2$ is true and $S_2 \Rightarrow S_1$ is true
- Simple recursive process evaluates an sentence.
 - $\neg P_{12} \wedge (P_{22} \vee P_{31})$

D. 模型、可满足性、有效性



语句的模型

- use the term **model** in place of “world”.
- We say **m is a model of a sentence α** (or m satisfies α) if **α is true in m** .
- **$M(\alpha)$** is the set of all models of α

- m is a model of α
- α is true in m

KB的模型

- A KB is a set of sentence, for instance
 $KB = \{P, P \vee R\}$
- A model **m is a model of KB** iff it is a model of all sentences in KB, that is, **all sentences in KB are true in m.**
- **$M(KB)$** is the set of all models of KB.

模型集合

- 给定问题描述所涉及的语句集合： $P, P \Rightarrow Q$ 。
请写出 $M(True), M(P)$

模型集合的运算

- $M(\alpha \wedge \beta) = M(\alpha) \cap M(\beta)$
- $M(\alpha \vee \beta) = M(\alpha) \cup M(\beta)$
- $M(\neg\alpha) = M(Truth) - M(\alpha)$

语句的有效性、可满足性

- α is *valid* if it is true in **all** models,
- α is *satisfiable* if it is true in **some** model
- α is *unsatisfiable* if it is true in **no** models
 - α is unsatisfiable, iff $\neg\alpha$ is valid
 - α is valid, iff $\neg\alpha$ is unsatisfiable

- m is a model of α
- α is true in m
- m satisfies α

KB的可满足性

- A KB is *satisfiable* iff it admits at least one model ($M(KB)$ is not empty); otherwise it is *unsatisfiable* ($M(KB)$ is empty)

E. 蕴涵与逻辑等价



蕴涵Entailment

- **Entailment** means that one thing follows from another:

$$KB \models \alpha$$

means

α follows from KB

- Knowledge base KB **entails** sentence α

$$KB \models \alpha$$

if and only if

α is true in all worlds where KB is true

$$KB \models \alpha \text{ 当且仅当 } M(KB) \subseteq M(\alpha)$$

$$KB \models \alpha: \text{ 若 } m \text{ 满足 } KB, \text{ 则 } m \text{ 也满足 } \alpha$$

逻辑等价

- 如果两语句 α 和 β 为真的模型集合相同，即 $M(\alpha) = M(\beta)$ ，则称它们是逻辑等价的，记为 $\alpha \equiv \beta$
- 任意两个语句 α 和 β 是等价的，当且仅当它们是互相蕴涵的： $\alpha \equiv \beta$ 当且仅当 $\alpha \models \beta$ 且 $\beta \models \alpha$

$$P \Rightarrow Q \equiv \neg P \vee Q$$

$$\alpha \wedge \text{True} \equiv \alpha$$

$$\alpha \wedge \text{False} \equiv \text{False}$$

$$\alpha \vee \text{True} \equiv \text{True}$$

$$\alpha \vee \text{False} \equiv \alpha$$

Wumpus世界的蕴涵

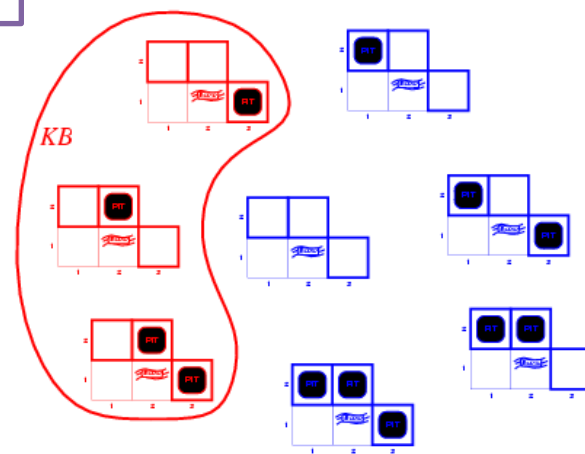
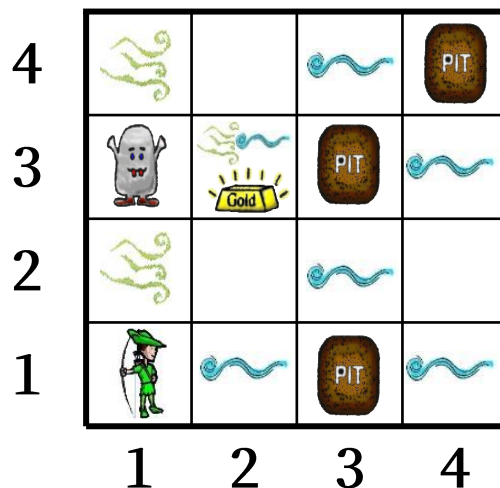
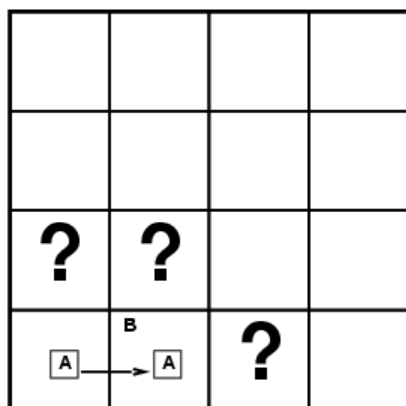
- Situation after detecting nothing in [1,1], moving right, breeze in [2,1]
- KB = wumpus-world rules + observations
- Consider possible models for KB assuming only pits

α_1 = “[1, 2]中没有无陷阱。”

α_2 = “[2, 2]中没有无陷阱。”

KB $\models \alpha_1$?

KB $\models \alpha_2$?



Wumpus世界的蕴涵

$\alpha_1 =$ “[1, 2]中没有无陷阱。”

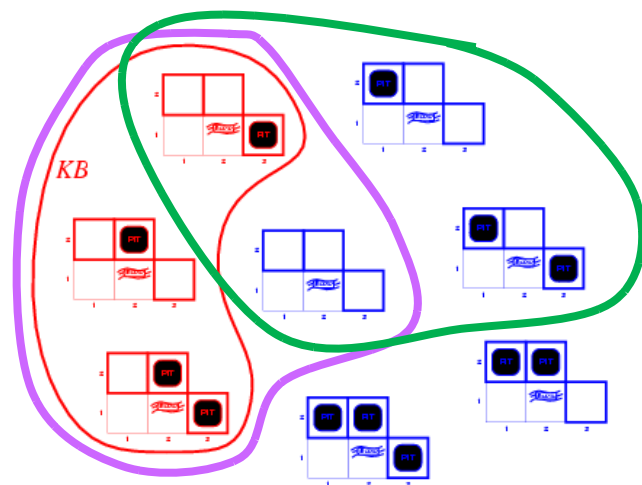
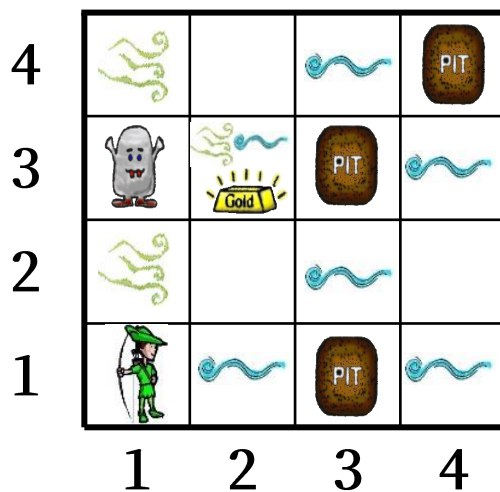
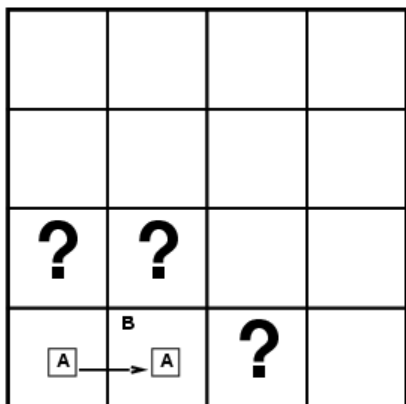
$\alpha_2 =$ “[2, 2]中没有无陷阱。”

$KB \models \alpha_1?$

$KB \models \alpha_2?$

$$M(KB) \subseteq M(\alpha_1)$$

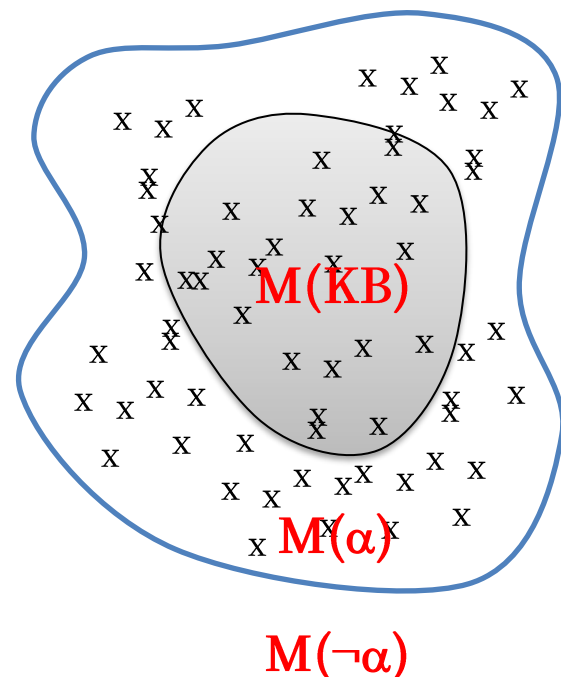
$$M(KB) \not\subseteq M(\alpha_2)$$



蕴涵： $KB \models \alpha$ 的等价形式

- $KB \models \alpha$
 - iff $M(KB) \text{ ______ } M(\alpha)$,
 - iff $KB \wedge \neg\alpha$ is ______,
 - iff $KB \Rightarrow \alpha$ is ______,
 - iff $KB, \neg\alpha \models \text{______}$

若 $KB \models \alpha$, 画出 $M(KB)$ 、 $M(\alpha)$ 、 $M(\neg\alpha)$ 的关系图



蕴涵的几种等价形式

蕴涵： $KB \models \alpha$ 的等价形式

□ $KB \models \alpha$

■ 根据蕴涵的定义有： $M(KB) \subseteq M(\alpha)$

■ 这等价于 $M(KB) \cap M(\neg\alpha) = \emptyset$,

即 $M(KB \wedge \neg\alpha) = \emptyset$,

因此 $KB \wedge \neg\alpha$ 是不可满足的

■ 也即 $\neg(KB \wedge \neg\alpha)$ 是有效的
等价于 $KB \Rightarrow \alpha$ 是有效的

■ $KB \wedge \neg\alpha$

$\equiv KB \wedge \neg\alpha \wedge \text{True}$

$\equiv KB \wedge \neg\alpha \wedge \neg \text{False}$

是不可满足的

这等价于 $KB \wedge \neg\alpha \models \text{False}$

小结

语法：为合法语句给出规范。

语义：定义每个语句在每个可能世界的真值。

m 是 α 的一个模型； α 在 m 下为真； m 满足 α 。

α 是可满足的：存在一个 m 满足 α 。

iff $M(\alpha)$ 不为空

α 是不可满足的：不存在 m 满足 α 。

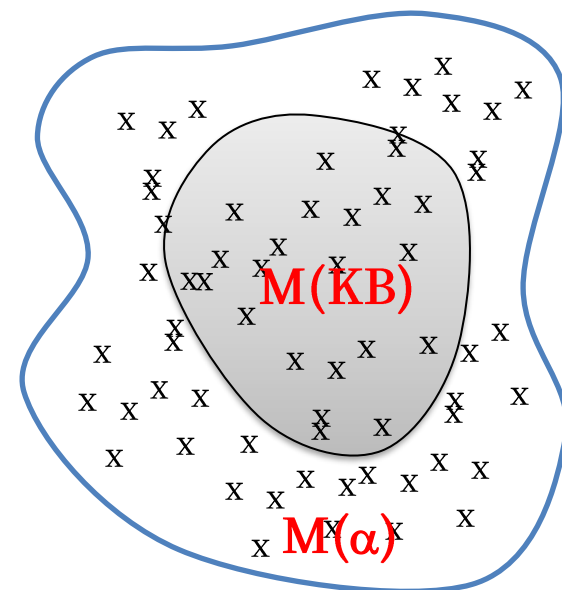
iff $M(\alpha)$ 为空

iff $\neg\alpha$ 是有效的

α 是有效的：所有模型都满足 α 。

iff $M(\neg\alpha)$ 为空

iff $\neg\alpha$ 不可满足的



$M(\neg\alpha)$

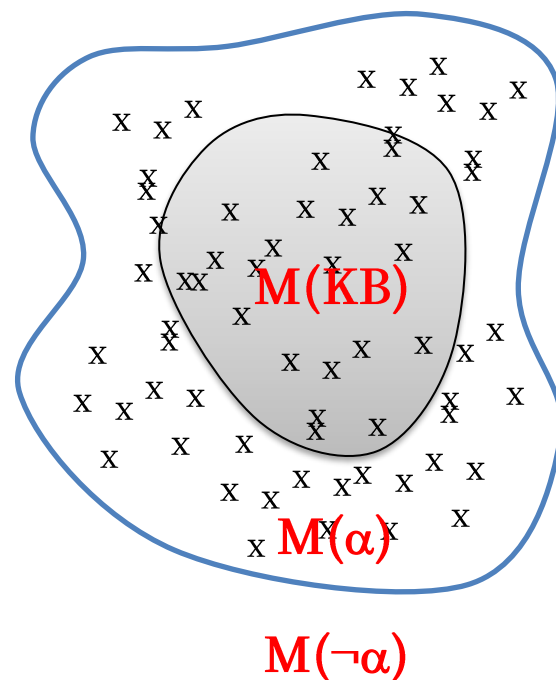
蕴涵： $KB \models \alpha$ 的等价形式

□ $KB \models \alpha$

- iff $M(KB) \subseteq M(\alpha)$,
- iff $KB \wedge \neg\alpha$ is **unsatisfiable**,
- iff $KB \Rightarrow \alpha$ is **valid**,
- iff $KB, \neg\alpha \models \text{False}$

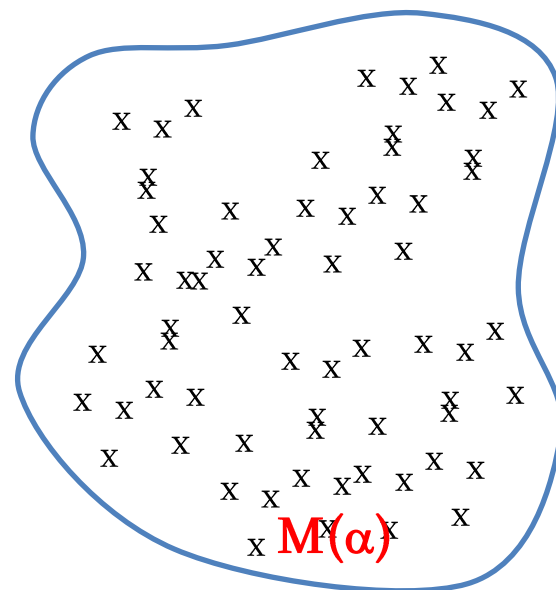
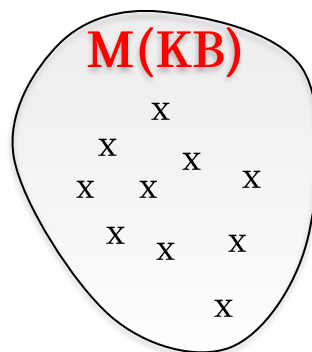
基于模型
基于可满足性
基于有效性
基于反演推理

$KB \models \alpha$: 在 KB 为真的
所有模型中, α 都为真



蕴涵： $KB \models \neg\alpha$ 的情况

□ $KB \models \neg\alpha$



$KB \models \alpha$: 在 KB 为真的所有模型中, α 都为真

$M(\neg\alpha)$

F. 逻辑推理



逻辑推理

- “Wumpus世界的蕴涵”中给出了如何用蕴涵推理出结论的实例。

$\alpha_1 =$ “[1, 2]中没有无陷阱。”
 $\alpha_2 =$ “[2, 2]中没有无陷阱。”
 $KB \models \alpha_1 ?$
 $KB \models \alpha_2 ?$

- 其中使用的推理算法为模型检验: 枚举KB为真的所有模型, 检验在这些模型下 α 是否都为真。即判断 $M(KB) \subseteq M(\alpha)$ 是否成立, 如果成立就有 $KB \models \alpha$, α 是从KB推理出的结论。

推理算法的可靠性和完备性

- $KB \vdash_i \alpha$: 推理算法i可以从KB推理出语句 α .
- *Soundness*(可靠性): 对于推理算法i从KB推理出的每条语句 α , 都有 $KB \models \alpha$, 则i是可靠的
- *Completeness*(完备性): 对于KB蕴涵的每条语句 α , 推理算法i都能从KB推理出 α , 则i是完备的
- TT-Entails?(KB, α)是可靠的、完备的
真值表枚举算法

Thanks